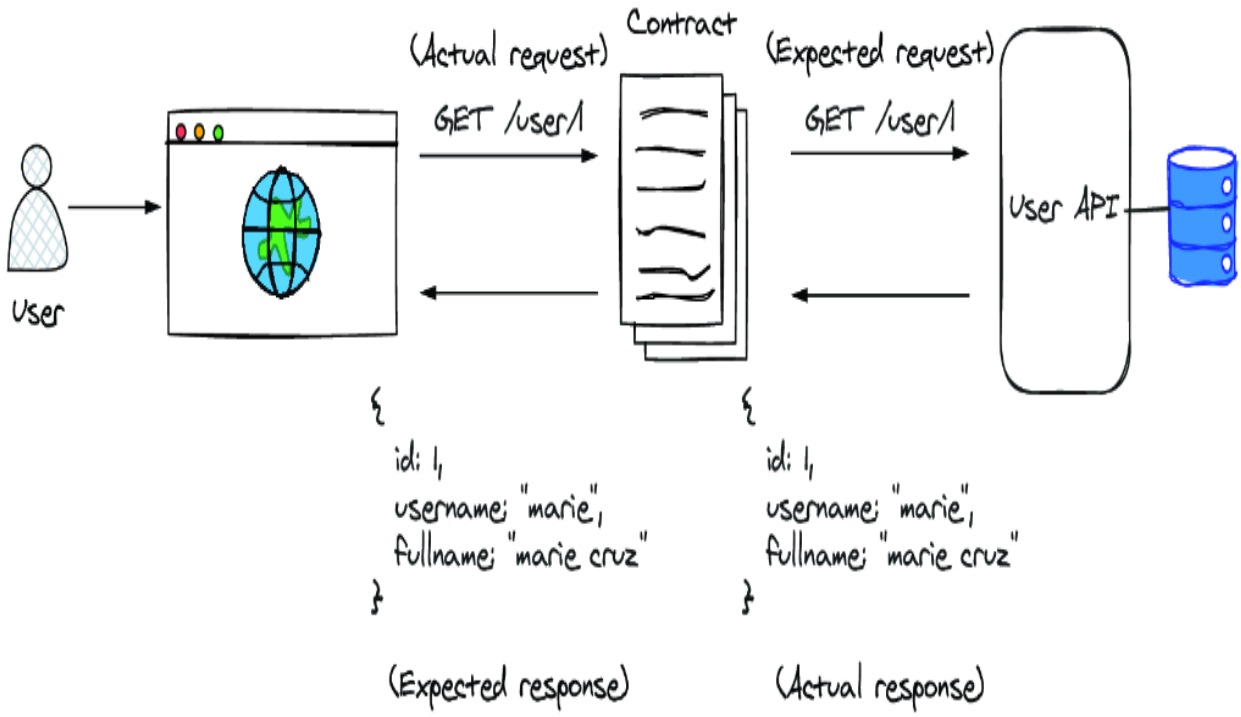


Figure 1.2 A web application and User API communicating via a contract, instead of directly with each other

The contract is in a JSON file containing the names of the two interacting systems: in this case, the web application and the User API. The contract also lists all interactions between these two systems, which we'll discuss further in chapter 3. The web application records the actual request and the response expected from the User API in the shared contract, which is uploaded to a shared broker that the User API can access directly. The User API pulls down the contract from the shared broker and replays the expected request from the web application. The contract testing tool will compare the User API's actual response to the expected response the web application requires.

1.2 Why contract testing matters

In an ideal world, there should be no breaking changes between two systems. However, we know that these scenarios can happen unfortunately. For example, suppose the User API changes the field from `fullName` to `name` and deploys this change to production. In that case, the two systems are now incompatible because the web application is expecting `fullName`, as shown in figure 1.3.